

James Barry

(410) 231-1200 | james.j.barry@icloud.com | College Park, MD
LinkedIn: James-J-Barry | GitHub: James-J-Barry | Portfolio: JimB.is

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, scikit-learn, LLM Prompt Engineering, Vector Embeddings, NLP, Quantum Machine Learning
Languages: Python, Java, JavaScript, TypeScript, C, R, MATLAB, Rust
Frameworks & Tools: React, Next.js, Node.js, AWS, Docker, PostgreSQL, Git, CI/CD, Linux, HPC (PBS/Slurm)

EDUCATION

University of Maryland

College Park, MD

B.S. Computer Science, Minor: Computational Finance GPA: 3.97/4.0

Expected Graduation May 2028

- **Coursework:** Machine Learning, Algorithms, Advanced Data Structures, Organization of Programming Languages, Computer Systems, Linear Algebra, Applied Probability & Statistics, Discrete Structures, Data Science, Business Value Chain, Financial Markets & Financial Datasets
- **Leadership:** Logistics Director, Bitcamp Hackathon: led ~20 organizers coordinating venue, hardware, and utilities for 1,400+ attendees; Education Chair, Kappa Theta Pi: ran professional development programming and corporate networking events for 70+ members

PROFESSIONAL EXPERIENCE

T. Rowe Price

Owings Mills, MD

Software Engineering Intern — AI Engineering & Application Development

May 2026 – Aug 2026

- Engineered a data-enrichment pipeline for an AI-powered AWS cost-optimization tool, surfacing **\$139,634/year** in savings across 517 EC2 instances by integrating pricing lookups and automated recommendation logic in a React/TypeScript frontend.
- Diagnosed a critical infrastructure failure by tracing a 58MB error response to a Splunk proxy misconfiguration, restoring access to **104,205 AWS resources** across 200 accounts by migrating 5 Python API modules.
- Optimized data-pipeline query architecture, increasing EC2 coverage from 515 to **4,154 instances** and uncovering an additional \$111,622/year in savings by resolving Splunk API result-cap limitations and data dilution in mixed-resource queries.
- Raised AI-generated report data quality from **76% to 96%** field coverage via a systematic audit of 79 fields across 15 panels.

Parsons Corp. @ ARL DoD Supercomputing Research Center

Aberdeen Proving Ground, MD

Software Engineer & Data Scientist Intern

Jun 2025 – Aug 2025

- Built and deployed a scalable evaluation system for semantic similarity algorithms on **100,000+ Army Test Incident Reports**, improving evaluator accuracy and operational efficiency.
- **Improved similarity search precision by over 280%** using multi-stage LLM summarization pipelines with advanced prompt engineering and JSON parsing.
- **Cut processing time by 8×** by parallelizing TIR LLM analysis across 8-node HPC clusters using PBS batch jobs.

First Year Innovation and Research Experience (FIRE)

College Park, MD

Quantum Machine Learning Researcher

Aug 2024 – Dec 2025

- Engineered a Quantum Principal Component Analysis (QPCA) algorithm using Quantum Phase Estimation (QPE) to extract eigenvalues without constructing full covariance matrices.
- Validated **86% reconstruction parity** with classical PCA on 16×16 MNIST digit images using Mean Squared Error.
- Demonstrated a complexity reduction from cubic to polylogarithmic time ($\mathcal{O}(N^3) \rightarrow \mathcal{O}(\text{poly}(\log N))$) — an exponential theoretical speedup; simulated density matrix exponentiation for IonQ hardware deployment.

PROJECTS

BarryBudget — Personal Finance Web & Mobile App (solo)

barrybudget.com

Next.js, React, TypeScript, PostgreSQL/Prisma, Plaid, Stripe

- Built a full-stack fintech platform spanning a Next.js web app, Prisma/PostgreSQL data layer, and React Native/Expo mobile app for bank aggregation, budgeting, and net-worth tracking.
- Integrated Plaid for live bank data with cursor-based transaction sync and AES-256-GCM token encryption; shipped usage-based Stripe billing with webhook-synced entitlements.
- Directed an AI-agentic development workflow (Claude Code) with a living technical spec and a GitHub Actions CI pipeline (Vitest, Playwright) gating every change.

Volatility Surface & Model-Divergence Engine

GitHub

Python, NumPy/SciPy, scikit-learn, Streamlit

- Engineered a live data pipeline ingesting and validating 1,000+ real-time market data records, feeding a Monte Carlo (GBM) simulation and a Newton-Raphson/Brent solver to recover latent model parameters.
- Cross-validated a constrained parametric fit against a gradient-boosted regression model of the live market surface, cutting model-vs-market error by **~90%** versus a naive baseline.